

	Case Name: Investment Product (2)	Sector	IT	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule Risk Analysis	Schedule with resources Schedule with costs Random simulation One of nine std. scenarios User defined distributions	
Submitted by	Matilde Casado and Margarida Antunes			
Date	2015			
File Name	C2015-23 Investment Product (2)		Project Control	Automatic tracking Tracking based on user input

1. Project description

Project authenticity

A detailed project focused on developing and implementing an Investment Product with tasks including design, validation, development, ICT testing, and post-go-live support.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	40	Serial/Parallel (SP)	17%
Planned Duration (PD)	122 days*	Activity Distribution (AD)	53%
Budget At Completion (BAC)	46.920 €	Length of Arcs (LA)	5%
Renewable Resources	16	Topological Float (TF)	39%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity				Time sensitivity		
	avg [%]	std dev [%]	skew [-]		avg [%]	std dev [%]	skew [-]
CRI-r	11.6	11.4	-1.4	CI	19.3	31.0	-1.4
CRI-rho	20.1	17.5	-0.9	SI	30.6	32.4	-0.9
CRI-tau	27.1	35.6	-1.5	SSI	6.2	12.1	-2.0
				CRI-r	10.9	13.4	-1.9
				CRI-rho	19.7	18.9	-0.9
				CRI-tau	29.7	34.2	-1.5

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	6.9	11.1	-1.9
CRI-rho	6.9	11.1	-1.9
CRI-tau	6.9	11.1	-1.9

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	101.1	-54.4	1	1.5	0.5
PV - SPI	46.3	46.3	CPI	1.8	0.8
PV - SCI	47.2	46.9	SPI	15.5	15.5
ED - 1	576.,4	398.3	SPI(t)	15.9	15.9
ED - SPI	46.3	46.3	SCI	15.5	15.5
ED - SCI	46.8	46.6	SCI(t)	15.9	15.9
ES - 1	105.2	-104.8	0.8 CPI + 0.2 SPI	11.2	11.2
ES - SPI(t)	23.9	15.1	0.8 CPI + 0.2 SPI(t)	12.2	12.2
ES - SCI(t)	24.1	15.2			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 4 tracking periods with a length of approximately 1 month. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	8724.8	-2506.5	-115.8	1.4	0.9	0.7	0.8
std dev	7257.6	3672.6	186.0	0.3	0.1	0.5	0.1
final	1411.5	900.0	12.0	1.4	1.0	1.0	1.0

2.3.3.2. Time forecasting

PD	122 days
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Real Duration	120 days
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Early	-1.6%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	124.3	1.0	0.7	-0.7
PV - SPI	124.8	0.5	0.8	-0.8
PV - SCI	125.3	0.5	0.9	-0.9
ED - 1	125.8	1.0	1.0	-1.0
ED - SPI	126.5	1.3	1.1	-1.1
ED - SCI	127.5	1.3	1.3	-1.3
ES - 1	128.5	1.3	1.5	-1.5
ES - SPI(t)	129.3	1.0	1.7	-1.7
ES - SCI(t)	129.8	0.5	1.8	-1.8

2.3.3.3. Cost forecasting

BAC	46.920 €
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Real Cost	32.805 €
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Under Budget	-43.0%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	38195.2	7257.6	4.1	-4.1
CPI	35216.2	7884.1	1.8	-1.8
SPI	44652.8	14225.9	9.0	-9.0
SPI(t)	312060.6	551652.5	212.8	-212.8
SCI	40463.1	12723.2	5.8	-5.8
SCI(t)	308837.0	553775.1	210.4	-210.4
0.8 CPI + 0.2 SPI	36375.2	8416.6	2.7	-2.7
0.8 CPI + 0.2 SPI(t)	38357.3	12834.1	4.2	-4.2